

Test Plan

1. Document Control

- **Project Name:** TfL “What-If” Simulation
- **Version:** 0.2
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- **Status:** Draft

2. Introduction

2.1 Purpose

This document describes the testing strategy, scope, and approach for the project. It defines how testing will be planned, executed, and managed throughout the software development lifecycle.

2.2 Scope

This test plan applies to all features derived from approved user stories. At this early stage, the focus is on validating requirements, planning test coverage, and defining processes rather than detailed test cases.

3. References

- Project Requirements / User Stories
- GitLab Repository
- University / Industry Software Engineering Standards

4. Test Strategy

4.1 Testing Levels

- **Unit Testing:** Developer-written tests for individual functions/ modules.
- **Integration Testing:** Verifying interactions between components.
- **System Testing:** End-to-end testing against user stories.
- **User Acceptance Testing:** Validation against acceptance criteria.
- **User Interface Testing:** Validating that the user interface works correctly.

4.2 Test Types

- Functional Testing
- Regression Testing
- Smoke Testing

- (Optional later) Performance & Security Testing

4.3 Test Approach

Testing will be risk-based and aligned with user stories. Each user story will include acceptance criteria that directly map to test cases.

5. Test Items

The following components will be tested:

- Core application features
- Authentication & authorisation (Only if implemented)
- Data validation and error handling

6. Test Environment

- **Version Control:** GitLab
- **Test Frameworks:** (JUnit, PyTest, Jest)
- **Browsers / Platforms:** Neo4j, React, Vite.
- **CI Pipeline:** GitLab CI

7. Entry & Exit Criteria

7.1 Entry Criteria

- User stories are approved
- Acceptance criteria defined
- Code committed to repository

7.2 Exit Criteria

- All planned tests executed
- No critical defects open
- Test summary report completed

8. Test Deliverables

- Test Plan
- Test Cases
- Automated Test Scripts
- Test Reports

9. Roles & Responsibilities

- **Testing Lead:** Test planning, coordination, reporting, Unit testing

- **Document Lead:** Structure, Consistency, Version Control, Documentation Quality
- **Project Management Lead:** Planning, coordinating tasks, tracking progress, delivered on time
- **Software Development Lead:** Implementation, code quality,

10. Defect Management

- Defects logged in GitLab Issues
- Each defect includes steps to reproduce, severity, and screenshots/logs

11. Risks & Mitigation

Risk	Impact	Mitigation
Unclear requirements	High	Early review of user stories
Limited Time	Medium	Prioritised Testing